

AMMAR HASSAN

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EDUCATION

COMSATS University, Lahore

Bachelor of Science in Software Engineering

2021 – 2026

Lahore, Pakistan

International School Lahore, DHA Campus

A-Levels (Advanced Level)

2019 – 2021

Lahore, Pakistan

EXPERIENCE

Corporate Events Company (Client)

Jul 2026 – Present

Freelance Web Developer

Lahore, Pakistan

- Rebuilding their **multi-page marketing site** end to end, working directly with the client through requirements gathering, design review rounds and revisions.
- Building it in **Astro** and deploying on **Cloudflare Pages**, with a shared motion system, SEO-targeted service pages and layouts verified at desktop and mobile breakpoints.

Punjab Safe Cities Authority (PSCA) 🔗

Jun 2024 – Aug 2024

Web Development Intern

Lahore, Pakistan

- Supported maintenance of **citizen-facing government portals**, working on **Node.js** services behind high-traffic provincial systems.
- Traced the data pipeline of Punjab's **Automated E-Challan Network** end to end, from AI-based camera detection through to web-dashboard visualization.

PROJECTS

CUI Central – Campus Ecosystem 🔗 | *Next.js, Socket.IO, Gemini AI, MongoDB*

- Architected a multi-tenant campus ERP consolidating **7+ core services** (Digital Cafeteria, Job Portal, Room Booking) into a single **Next.js 15** monorepo. Implemented **RBAC** (Role-Based Access Control) for Admin, Faculty and Student workflows.
- Engineered a real-time layer with **Socket.IO** for live faculty availability and order status updates. Integrated a **RAG-based AI agent** (Google Gemini) with MongoDB Vector Search to handle student support queries.
- [Live Site](#)

Cloud Visitor Counter – Serverless Backend 🔗 | *AWS, Terraform, Python, GitHub Actions*

- Built a serverless backend tracking live visitor counts on AWS (Lambda, API Gateway, DynamoDB), with all infrastructure provisioned in **Terraform** (IaC).
- Automated deployment through **GitHub Actions** with keyless **OIDC** authentication, with atomic counter increments and remote state management.
- [GitHub](#)

Check! – Multiplayer Game 🔗 | *Next.js, Node.js, Socket.IO, XState*

- Designed a custom WebSocket game engine supporting concurrent rooms, holding **<50ms latency** for game state sync over event-driven **Socket.IO**. Used **XState** state machines to manage game logic and remove race conditions across multiple clients.
- [Live Site](#) | [GitHub](#)

Karting Analysis – Telemetry Pipeline 🔗 | *Python, Pandas, NumPy, Matplotlib*

- Built a Python pipeline to ingest and clean **50,000+ telemetry points**, cutting processing time by **30%** by replacing iterative loops with **NumPy** vectorization. Generated statistical heatmaps to visualize driver consistency and degradation.
- [Live Site](#) | [GitHub](#)

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, C++, SQL

Web & Frameworks: Next.js, React, Node.js, Express, Socket.IO, Tailwind CSS

Data & AI: Pandas, NumPy, Matplotlib, RAG, Vector Search, Gemini API

Cloud & Tools: AWS, Terraform, Docker, Git, GitHub Actions, MongoDB, PostgreSQL, Linux